FINAL REPORT

BIODEGRADABILITY TEST AS PER OECD 301B modified

FOR TRANSOL SYNTH 100

SUBMITTED BY

SAVITA POLYMERS LTD

ENVIRONMENTAL DIVISION LABORATORY, MUMBAI

INTERTEK INDIA PRIVATE LIMITED

Client: SAVITA POLYMERS LTD

Sample registration date: 10/03/2022

Analysis starting date: 10/03/2022 (Preconditioning) **Analysis completed on:** 28/04/2022

Name of product: TRANSOL SYNTH 100

Quantity received and packing: ALUMINIUM PACKET and the quantity is 200gms

Sample details: TRANSOL SYNTH 100

Test Required: Biodegradability test as per **OECD 301B**

Sampling done by: Sample not drawn by Intertek

Report No: MUM/000752/2022

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LABORATORY

Testing as presented in this report was conducted by Environmental division of Intertek India Private Limited. The testing facility is located at F wing, 2nd Floor, Chandivali Saki vihar Road, Andheri (East), Mumbai – 400 072, India.

SAMPLE RECEIPT

The sample was received on 10/03/2022 at the Intertek testing facility. The package was received by courier. Samples were at ambient temperature in good condition with no evidence of damage or contamination.

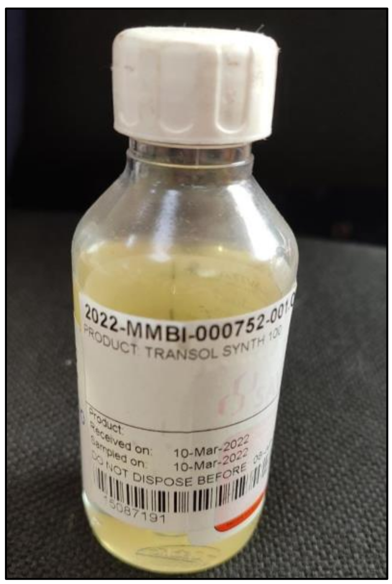

Figure 1: Test Sample

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PROJECT DESCRIPTION:

TRANSOL SYNTH 100 sample was submitted by **SAVITA POLYMERS LTD** for studying the Readily biodegradability as per OECD 301 B. A measured volume of inoculated mineral medium, containing a known concentration of the test substance (10-20 mg DOC or TOC/l) as the nominal sole source of organic carbon is aerated by the passage of carbon dioxide-free air at a controlled rate in the dark or in diffuse light. Degradation is followed over 28 days by determining the carbon dioxide produced. The CO_2 is trapped in barium or sodium hydroxide and is measured by titration of the residual hydroxide or as inorganic carbon. The amount of carbon dioxide produced from the test substance (corrected for that derived from the blank inoculum) is expressed as a percentage of $ThCO_2$. The degree of biodegradation may also be calculated from supplemental DOC analysis made at the beginning and end of incubation. OECD 301B biodegradability testing monitors the degree of activity of microorganisms exposed to a material that is being tested for a biodegradable status. In the test, if the microorganisms recognize the material as a food source, then an increase in biological activity is observed through data collection specifically designed to assess biological conversion of organic carbon to inorganic carbon. If the material is not a recognizable food source or is toxic or inhibitory, then there is no measurable increase in biological activity or, in some cases, there is a marked decrease in activity relative to a biodegradable control.

INOCULUM COLLECTION AND CONDITIONING

The inoculum was derived from Sewage effluent source. Following collection, the inoculum was immediately transported to the testing facility and was aerated with CO_2 free air. Pre-conditioning was done by aerating the effluent for 5-7 days at the test temperature.

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TEST SETUP

The test vessels were prepared and analyzed for the study: Blank Control, Test Products, and Positive Control. Reactors were setup in duplicates for statistical validation of results.

The Test Products and Positive Control are analyzed for total organic carbon prior to study initiation to determine appropriate dilutions that yield a concentration of 10-20 mg//L per method requirements. Treatments included test flasks containing test substance and inoculum (test suspension); only inoculum (inoculum blank); a standard reference material, known to be biodegradable, was selected as reference and positive control.

TEST RESULTS:

Based on the testing conducted in accordance with the specified method, the test sample **TRANSOL SYNTH 100** sample achieved 92.91 % biodegradation in relative to reference control after 28 days of testing (Table 1).

Table 1: Represent % biodegradation of Positive control , Reference Control & the Test sample.

Time (Day)	% Biodegradation		
	Positive control Aniline	Reference control with ref. to product type (Rapeseed Oil)	Test Sample
0	0.00	0.00	0.00
3	21.37	25.26	20.07
5	32.77	49.11	29.73
7	48.45	64.54	41.62
10	72.67	74.36	69.12
14	78.37	87.55	75.56
18	86.92	94.01	83.00
21	91.19	96.81	85.47
24	92.62	97.65	91.67
28	94.61	100.00	92.91

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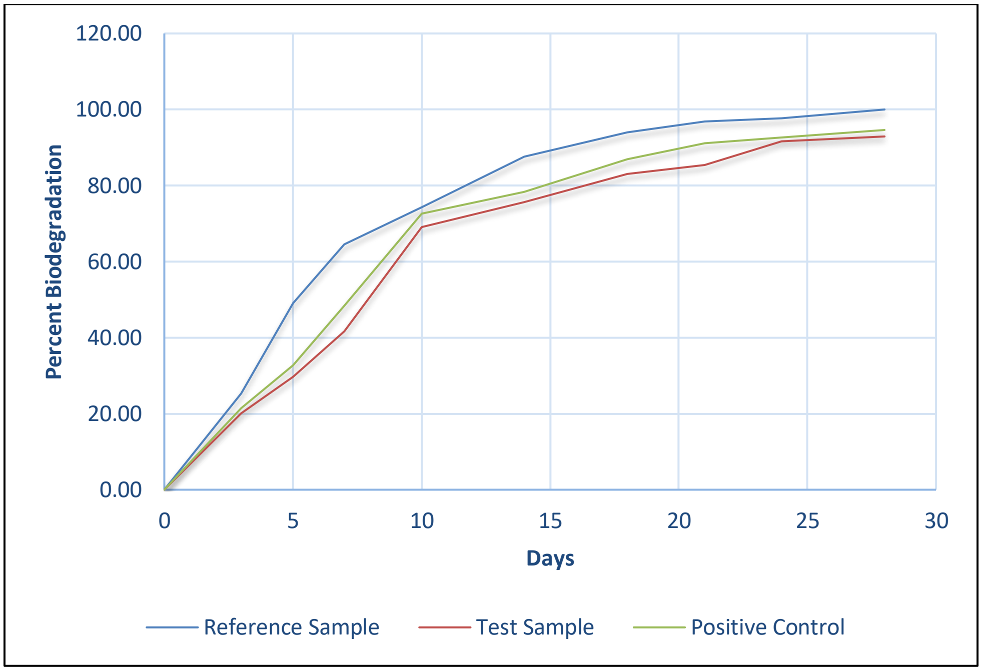

Graph 1: Graphical representation of biodegradability tests OECD 301B on TRANSOL SYNTH 100 sample.

The **TRANSOL SYNTH 100** sample demonstrated biodegradability by achieving percentage biodegradation of 92.91 % in relative to reference control (Rapeseed Oil) which showed 100 % by day 28. **TRANSOL SYNTH 100** sample demonstrated ready biodegradability according to the conditions set forth in OECD 301B requirements of achieving the 60% conversion to CO2 within the required timeframe.

The microscopic examination of the test sample was done after 28 days of Biodegradability study. Microbial Biomass residue was observed in huge amount.

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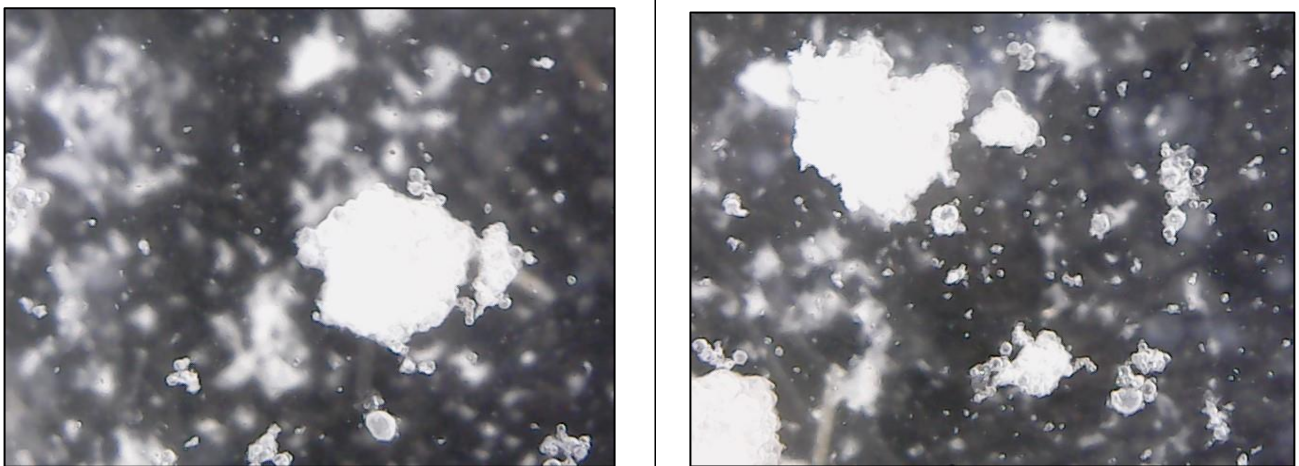

Figure 2: Microscopic examination of the test samples after biodegradation

Conclusion

Test sample **TRANSOL SYNTH 100** is readily biodegradable in accordance to the conditions set forth in OECD 301B requirements of achieving the 60% conversion to CO_2 within the required timeframe.

--- End of Report ------------------------------

Authorized Signatory

Jayashree

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Jayashree Acharya
Assistant Manager- Environment laboratory

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